IBM FileNet P8 V5.5.8

Step-by-Step Hands-On Implementation Guide

Day 7 – Workflow Development using Process Designer

Process Designer Setup | Build Approval Workflow | Deploy & Test

Author	Course Day	Labs	Duration
Rajesh Pasham	**Day 7 of 10**	**3 Labs**	**~3.5 Hours**

LAB-07-01	LAB-07-02	LAB-07-03
Install & Connect Process Designer	**Build Document Approval Workflow**	**Deploy, Test & Verify Workflow**
40 min | IBM IM + PE Connection	*90 min | Full 11-step workflow map*	*40 min | Deploy to PE + live test*

Prerequisites & Environment State

Environment State Before Day 7 Labs

Day 7 builds directly on the Process Engine installed and configured in Day 6. Verify the following before starting:

PE Service	IBM FileNet Process Engine service is Running in services.msc
PE Port	Port 32768 is listening: netstat -an | findstr 32768 shows LISTENING
PE Connection Point	PE_ConnectionPoint is Online in ACCE > P8Domain > Process Engine > Connection Points
TrainingRoster	TrainingRoster exists with TrainingApprovalQueue and TrainingReviewQueue
CE/WAS Running	Content Engine running on WAS. ACCE accessible at http://localhost:9080/acce
Object Store	TrainingOS exists with at least one document for workflow testing
Process Designer Software	ProcessDesigner_5.5.8_WIN.zip available in software repository
IBM IM Available	IBM Installation Manager V1.9+ is installed
Working Directory	C:\Day7Labs\ will be created to store workflow .pep files

LAB-07-01 | Install & Connect IBM Process Designer V5.5.8

Duration: 40 Minutes | Install Process Designer and establish connection to the Process Engine

NOTE: START STATE: IBM FileNet Process Engine V5.5.8 is installed and running (port 32768). PE_ConnectionPoint is Online in ACCE. IBM Installation Manager is installed on this Windows server.

Objective

In this lab you will:

- Create the Day 7 working directory for workflow .pep files
- Install IBM Process Designer V5.5.8 using IBM Installation Manager
- Launch Process Designer and configure the PE server connection
- Test the connection and verify Process Designer can see TrainingRoster and queues

STEP 1 | Create Working Directory and Verify PE is Running | 5 min

1. Open a Command Prompt as Administrator and create the working directory:

```
mkdir C:\Day7Labs
echo Working directory created
```

2. Verify PE is running:

```
netstat -an | findstr 32768
# Expected: TCP  0.0.0.0:32768  0.0.0.0:0  LISTENING

# If not listening, start the PE service:
net start "IBM FileNet Process Engine"
```

3. Open ACCE and verify PE_ConnectionPoint is Online:

ACCE URL	`http://localhost:9080/acce`
Navigate to	`P8Domain > Process Engine > Connection Points`
Expected Status	`PE_ConnectionPoint — Online (green indicator)`

	VERIFICATION CHECKLIST
[ ]	Port 32768 shows LISTENING
[ ]	PE_ConnectionPoint shows Online in ACCE

STEP 2 | Install IBM Process Designer V5.5.8 via IBM IM | 20 min

4. Launch IBM Installation Manager: Start Menu > IBM Installation Manager
5. Click File > Preferences > Repositories > Add Repository
6. Browse to the Process Designer repository: C:\Software\PD\repository.config
7. Click OK > Apply > OK
8. Click Install from the IBM IM main screen
9. IBM IM discovers IBM FileNet Process Designer V5.5.8 in the added repository

10. Check the checkbox next to Process Designer 5.5.8 and click Next
11. Accept the license agreement and click Next
12. Set the installation directory:

Installation Directory	`C:\IBM\FileNet\ProcessDesigner`
Shared Resources	`C:\IBM\IMShared (existing from Day 2)`

13. Click Next > review summary > click Install
14. Wait approximately 10 minutes for installation to complete
15. Click Finish when the installation completes

> **TIP:** If Process Designer does not appear in the IBM IM repository list, verify the repository.config file path. The file must be inside the extracted Process Designer software folder (not the ZIP itself).

	VERIFICATION CHECKLIST
[ ]	IBM IM installation completes with: Process Designer installed successfully
[ ]	Directory C:\IBM\FileNet\ProcessDesigner\ exists
[ ]	ProcessDesigner.exe exists at C:\IBM\FileNet\ProcessDesigner\ProcessDesigner.exe

STEP 3 — Launch Process Designer and Configure PE Connection (10 min)

16. Launch Process Designer: double-click C:\IBM\FileNet\ProcessDesigner\ProcessDesigner.exe
17. Process Designer opens with the main canvas and step palette
18. Click File > Connection > New Connection
19. Fill in the connection details:

Connection Name	`TrainingPEConnection`
Server	`localhost`
Port	`32768`
User Name	`ceadmin`
Password	`ceadmin`
Default Roster	`TrainingRoster`

20. Click Test Connection
21. Expected result: Connection successful — Process Designer connects to PE and retrieves roster/queue list

> **WARNING:** If Test Connection fails: (1) Verify PE service is Running in services.msc, (2) Verify netstat shows port 32768 LISTENING, (3) Verify ceadmin credentials are correct, (4) Check Windows Firewall is not blocking port 32768.

22. Click OK to save the connection
23. In Process Designer, click File > Connection > TrainingPEConnection to activate it

[SCREENSHOT]
IBM Process Designer main window with TrainingPEConnection active, showing TrainingRoster in the connection dropdown

	VERIFICATION CHECKLIST
[ ]	Process Designer launches without errors
[ ]	Test Connection result: Connection successful
[ ]	TrainingPEConnection saved and active
[ ]	TrainingRoster and queues visible via File > Connection

STEP 4 — Explore Process Designer Interface (5 min)

Before building the workflow, familiarise yourself with the Process Designer interface:

Left Panel – Step Palette	Lists all available step types. Drag steps from here to the canvas.
Center – Process Map Canvas	The main design area. Drop steps here, draw connectors between them.
Right Panel – Properties	Shows properties of the selected step or connector. Edit here.
Bottom Panel – Data Fields	Define workflow data fields for this map. Name, data type, default value.
Toolbar	Save, Open, Deploy, Validate, Print, Zoom controls.
Map menu	Map > Validate Map, Map > Deploy to Process Engine, Map > Properties.

LAB-07-02 | Build the Document Approval Workflow

Duration: 90 Minutes | Design a complete 11-step Document Approval Workflow in Process Designer

NOTE: START STATE: Lab-07-01 completed. Process Designer is running with TrainingPEConnection active. TrainingRoster, TrainingApprovalQueue, and TrainingReviewQueue exist on the PE server.

Objective

In this lab you will build DocumentApprovalWorkflow — a complete, deployable 5-stage document approval process map with data fields, queue steps, routing decisions, escalations, and a rejection loop.

Workflow Map Overview (What We Are Building)

The workflow implements the following business process:

```
STAGE 1 - SUBMISSION
  LaunchStep --> InitializeFieldsStep --> SubmissionNotifyStep

STAGE 2 - REVIEW
  TrainingReviewQueue --> ReviewDecision
  [If reviewer marks F_ReviewerComments = 'Pass'] --> STAGE 3
  [If reviewer marks F_ReviewerComments = 'Fail'] --> RevisionQueue --> back to
Review

STAGE 3 - APPROVAL
  TrainingApprovalQueue --> ApprovalDecision
  [If F_ApprovalStatus = 'Approved']  --> STAGE 4 (Notify + Complete)
  [If F_ApprovalStatus = 'Rejected']  --> RejectionCountCheck
                                        --> RevisionQueue (if count < 3)
                                        --> AutoRejectStep (if count >= 3)

STAGE 4 - COMPLETION
  ApprovalNotifyStep --> MilestoneStep --> TerminateStep
```

STEP 1 — Create New Process Map (5 min)

1. In Process Designer, click File > New Map
2. The Map Properties dialog appears. Fill in:

Map Name	`DocumentApprovalWorkflow`
Description	`4-stage document approval: Submit, Review, Approve/Reject, Complete`
Roster	`TrainingRoster`
Save Location	`C:\Day7Labs\DocumentApprovalWorkflow.pep`

3. Click OK. Process Designer creates a new blank canvas with a single Launch Step already placed.
4. Click File > Save As > C:\Day7Labs\DocumentApprovalWorkflow.pep

STEP 2 Define Workflow Data Fields — 10 min

Before placing steps, define all data fields. Click the Data Fields tab at the bottom of the screen.

5. Click Add Field and create each field with the following settings:

`F_Subject`	String	`(empty)`	YES	*System field — required at launch*
`F_BoundDocument`	String	`(empty)`	No	*Set by Process Router*
`F_Priority`	String	`Medium`	No	*High / Medium / Low*
`F_ReviewerComments`	String	`(empty)`	No	*Pass / Fail*
`F_ApprovalStatus`	String	`Pending`	No	*Pending / Approved / Rejected*
`F_ApprovedBy`	String	`(empty)`	No	*Set by approver*
`F_RejectionCount`	Integer	`0`	No	*Rejection loop counter*
`F_ReviewCount`	Integer	`0`	No	*Review loop counter*

6. Click Save (Ctrl+S) after adding all fields

	VERIFICATION CHECKLIST
[ ]	All 8 data fields created in the Data Fields panel
[ ]	F_Subject marked as Required
[ ]	F_ApprovalStatus has Initial Value: Pending
[ ]	F_RejectionCount has Initial Value: 0

STEP 3 Configure the Launch Step — 5 min

7. Double-click the Launch Step already on the canvas to open its Properties
8. Set the following properties:

Step Label	`Submit Document for Approval`
Instructions	`Enter the document subject and priority before submitting for approval.`
Required Fields	`F_Subject (drag from Data Fields panel to Required list)`
Visible Fields	`F_Subject, F_Priority (users see and set these at launch)`

9. Click OK to close the Launch Step properties

STEP 4 — Add Stage 1: Initialize Fields (System Step) — 8 min

10. From the Step Palette, drag a System Step onto the canvas, placing it to the right of the Launch Step
11. Double-click the System Step to open its Properties

Step Label	`InitializeFields`
Step Type	`Script Step (select Script from Step Type dropdown)`

12. In the Script tab, enter the following initialization script:

```
// Initialize all workflow data fields to default values
F_ApprovalStatus = "Pending";
F_RejectionCount = 0;
F_ReviewCount = 0;
F_ReviewerComments = "";
F_ApprovedBy = "";
```

13. Click OK
14. Draw an unconditional connector from LaunchStep to InitializeFields:

- Select the Connector tool from the toolbar (arrow icon)
- Click on the LaunchStep, then click on InitializeFields

	VERIFICATION CHECKLIST
[ ]	InitializeFields step placed on canvas
[ ]	Script entered correctly
[ ]	Unconditional connector drawn from LaunchStep to InitializeFields

STEP 5 — Add Stage 2: Review Queue Step — 10 min

15. Drag a Queue Step from the palette onto the canvas
16. Double-click the Queue Step to open its Properties:

Step Label	`Document Review Required`
Queue Name	`TrainingReviewQueue (select from dropdown)`
Instructions	`Review the attached document for completeness and quality. Set your comments to Pass or Fail.`
Visible Fields	`F_Subject, F_Priority, F_ReviewerComments, F_BoundDocument`
Required Fields	`F_ReviewerComments (reviewer must enter Pass or Fail)`

17. Click the Deadline / Escalation tab:

Deadline Expression	`Dates.add(Dates.currentDate(), 2, 'd')`

Escalation Type	`Reassign`
Reassign To	`ceadmin`
Escalation Delay	`4 hours after deadline`

18. Click OK. Draw an unconditional connector from InitializeFields to Document Review Required.

> **TIP:** The Deadline Expression Dates.add(Dates.currentDate(), 2, 'd') sets the step deadline to 2 days from when the work item enters the queue. If no reviewer acts within 2 days + 4 hours, PE automatically reassigns it to ceadmin.

STEP 6 — Add Stage 2: Review Routing Decision — 8 min

19. Drag a Routing Decision (diamond) from the palette onto the canvas, placing it after the Review Queue Step
20. Label the Routing Decision: ReviewOutcome
21. Draw an unconditional connector from Document Review Required to ReviewOutcome
22. Now add two conditional connectors out of ReviewOutcome:

Connector 1 – Review Passed:

- Draw connector from ReviewOutcome toward the right (toward Approval stage)
- Double-click the connector to set its routing expression:

```
F_ReviewerComments == "Pass"
```

- Label the connector: Review Passed

Connector 2 – Review Failed (Default):

- Draw a connector from ReviewOutcome back toward the TrainingReviewQueue (creating a loop)
- Set this connector as the Default connector (check 'Default' in connector properties)
- In a System Step before the Review Queue (add IncrementReviewCount step):

```
F_ReviewCount = F_ReviewCount + 1;
```

[SCREENSHOT]
Process Designer canvas showing LaunchStep, InitializeFields, ReviewQueue, ReviewOutcome diamond with two connectors

	VERIFICATION CHECKLIST
[ ]	ReviewOutcome routing decision placed and labeled
[ ]	Connector with F_ReviewerComments == 'Pass' expression drawn
[ ]	Default connector (review loop) drawn back to queue with IncrementReviewCount step

STEP 7 — Add Stage 3: Approval Queue Step — 10 min

23. Drag a Queue Step onto the canvas for the Approval stage (after the Review Passed connector)
24. Double-click to set properties:

Step Label	`Manager Approval Required`
Queue Name	`TrainingApprovalQueue (select from dropdown)`
Instructions	`Review the document and set Approval Status to Approved or Rejected. Add your username to Approved By if approving.`
Visible Fields	`F_Subject, F_Priority, F_ApprovalStatus, F_ApprovedBy, F_ReviewerComments, F_BoundDocument`
Required Fields	`F_ApprovalStatus, F_ApprovedBy`

25. Deadline / Escalation tab:

Deadline Expression	`Dates.add(Dates.currentDate(), 1, 'd')`
Escalation Type	`Reassign to ceadmin after 2 hours`

26. Click OK. Draw the 'Review Passed' connector to this Queue Step.

STEP 8 — Add Stage 3: Approval Routing Decision — 12 min

27. Drag another Routing Decision (diamond) onto the canvas after Manager Approval Required
28. Label it: ApprovalOutcome
29. Draw an unconditional connector from Manager Approval Required to ApprovalOutcome

Add three conditional connectors from ApprovalOutcome:

Connector A – Approved:

```
F_ApprovalStatus == "Approved"
```

- Label: Document Approved
- Routes to: Stage 4 (Notify step)

Connector B – Rejected and can retry (count < 3):

```
F_ApprovalStatus == "Rejected" && F_RejectionCount < 3
```

- Label: Rejected – Return for Revision
- Draw this connector to an IncrementRejection System Step:

```
F_RejectionCount = F_RejectionCount + 1;
```

- Then back to TrainingReviewQueue (rejection loop — document returns to reviewer)

Connector C – Rejected and max attempts reached (Default):

- Set as Default connector
- Routes to AutoRejectStep (System Step):

```
F_ApprovalStatus = "AutoRejected";
// Optional: set a milestone here
// Milestone: 'AutoRejectedAfterMaxAttempts'
```

- AutoRejectStep connects to TerminateStep

[SCREENSHOT]
Process Designer canvas showing full approval routing with ApprovalOutcome diamond and three connectors including rejection loop

	VERIFICATION CHECKLIST
[ ]	ApprovalOutcome routing decision placed with 3 connectors
[ ]	F_ApprovalStatus == 'Approved' connector labeled and routed to notify stage
[ ]	Rejection with count < 3 connector routes back to review queue
[ ]	Default connector routes to AutoRejectStep

STEP 9 — Add Stage 4: Notify and Complete (10 min)

30. After the Approved connector, add a Notify Step:

Step Label	`Approval Notification`
Step Type	`Notify Step`
Message Subject	`Document Approved: {F_Subject}`
Message Body	`Your document has been approved by {F_ApprovedBy} on {Dates.currentDate()}.`
Recipients	`F_Originator (the user who launched the workflow)`

31. After ApprovalNotification, add a Milestone Step:

Step Label	`WorkflowComplete`
Milestone Name	`ProcessComplete`

32. Finally, add a Terminate Step and connect it from both the Milestone Step and the AutoRejectStep:

- Drag Terminate from palette (circle icon)
- Draw connector from MilestoneStep to Terminate
- Draw connector from AutoRejectStep to Terminate

IMPORTANT: Every branch in the workflow MUST eventually reach the Terminate step. This is the most common validation error. Before saving, trace every possible path through the workflow and confirm it ends at Terminate.

STEP 10 — Validate the Workflow Map (5 min)

33. Press Ctrl+S to save the workflow
34. Click Map > Validate Map
35. The validation report appears. Review all items:

Severity	Message	Action Required

ERROR	`No Terminate step reachable from [step name]`	Add Terminate step and connect all orphaned branches
ERROR	`Invalid expression: [expression text]`	Fix the routing expression syntax
ERROR	`Queue 'XYZ' not found on PE server`	Verify queue name matches exactly (case-sensitive) in TrainingRoster
WARNING	`Unreachable step: [step name]`	The step has no incoming connector. Delete or connect it.
INFO	`Map validated successfully`	No action needed — proceed to deploy

36. Fix ALL errors before proceeding to Step 3 (Deploy). Warnings can be accepted if understood.

	VERIFICATION CHECKLIST
[ ]	Validation completes with: Map validated successfully (or only warnings, no errors)
[ ]	No ERROR severity items in the validation report
[ ]	DocumentApprovalWorkflow.pep saved at C:\Day7Labs\

LAB-07-03 | Deploy, Test & Verify the Workflow

Duration: 40 Minutes | Deploy DocumentApprovalWorkflow to PE and run a complete end-to-end test

NOTE: START STATE: Lab-07-02 completed. DocumentApprovalWorkflow.pep is saved at C:\Day7Labs\ and has passed validation. Process Designer is running with TrainingPEConnection active.

Objective

In this lab you will:

- Deploy the DocumentApprovalWorkflow to the FileNet P8 V5.5.8 Process Engine
- Verify the workflow definition appears in ACCE Process Configuration Console
- Create a test document in TrainingOS and manually launch the workflow
- Claim and step the work item through the TrainingReviewQueue
- Approve the document in TrainingApprovalQueue and verify workflow completes

STEP 1 — Deploy Workflow to Process Engine (10 in)

1. In Process Designer, ensure DocumentApprovalWorkflow.pep is open and validated
2. Click Map > Deploy to Process Engine
3. The Deploy dialog opens:

PE Connection	`TrainingPEConnection (localhost:32768)`
Deploy Mode	`New (first-time deployment)`
Map Name	`DocumentApprovalWorkflow (auto-populated)`

4. Click Deploy
5. Expected messages in the Deploy Log:

```
Connecting to Process Engine (localhost:32768)...
Connected successfully.
Uploading workflow definition: DocumentApprovalWorkflow...
Workflow definition deployed successfully.
Version 1 of DocumentApprovalWorkflow is now active.
```

TIP: If deployment fails with 'Workflow already exists', click Map > Deploy Options and choose 'Deploy as new version' or delete the existing definition from ACCE > Process Configuration Console > Workflow Definitions first.

[SCREENSHOT]
Process Designer Deploy Log showing: Workflow definition deployed successfully

	VERIFICATION CHECKLIST
[ ]	Deploy dialog shows: Workflow definition deployed successfully
[ ]	Version 1 of DocumentApprovalWorkflow is now active
[ ]	No error messages in the deploy log

STEP 2 — Verify Deployment in ACCE Process Configuration Console — 5 min

6. Switch to ACCE in the browser: http://localhost:9080/acce
7. Click Tools > Process Configuration Console
8. In the console, expand the left panel and navigate to Workflow Definitions
9. Verify DocumentApprovalWorkflow appears in the list with:

Workflow Name	`DocumentApprovalWorkflow`
Version	`1`
Status	`Active`
Roster	`TrainingRoster`

[SCREENSHOT]
ACCE Process Configuration Console showing DocumentApprovalWorkflow in Workflow Definitions list with Version 1 Active

	VERIFICATION CHECKLIST
[ ]	DocumentApprovalWorkflow visible in ACCE Process Configuration Console
[ ]	Status: Active, Version: 1

STEP 3 — Create a Test Document in TrainingOS — 5 min

10. In ACCE, navigate to: Object Stores > TrainingOS > Browse > Root Folder
11. If /API_Labs folder exists from Day 9 prep, use it. Otherwise:
12. Right-click Root Folder > New Document:

Document Title	`Workflow Test Document 01`
Class	`Document`
Content	`Upload any small test file (C:\Day7Labs\test.txt if available)`

13. Click Finish to create the document
14. Note the document ID (visible in Properties > Info tab). You will use this to launch the workflow.

STEP 4 — Manually Launch the Workflow on the Test Document — 8 min

15. In ACCE Browse, right-click the Workflow Test Document 01
16. Select Launch Workflow (or similar option in the right-click menu)
17. The Launch Workflow dialog opens:

Workflow Definition	`DocumentApprovalWorkflow`
Launch Step	`Submit Document for Approval`

F_Subject	`Workflow Test Document 01`
F_Priority	`High`

18. Click Launch
19. Expected: A workflow instance is created. Note the Work Object Number returned.

> **TIP:** If Launch Workflow is not available in the right-click menu, verify: (1) The document class has the WorkflowAutoLaunch property enabled, or (2) use ACCE > Process Configuration Console > Launch New Workflow directly.

	VERIFICATION CHECKLIST
[ ]	Workflow launched successfully with a Work Object Number
[ ]	Work Object Number noted for tracking

STEP 5 — Step the Workflow Through Review Queue (7 min)

20. In ACCE > Process Configuration Console, expand TrainingRoster > TrainingReviewQueue
21. A work item for 'Workflow Test Document 01' should appear in the queue depth (count: 1)
22. Click on the work item to view its details
23. To simulate a reviewer action, click Lock/Claim on the work item
24. Set F_ReviewerComments = Pass in the work item fields
25. Click Step (or Dispatch) to move the work item to the next step
26. The work item should disappear from TrainingReviewQueue and appear in TrainingApprovalQueue

[SCREENSHOT]
ACCE Process Configuration Console showing work item in TrainingApprovalQueue after review step

	VERIFICATION CHECKLIST
[ ]	Work item visible in TrainingReviewQueue with F_Subject: Workflow Test Document 01
[ ]	After stepping with F_ReviewerComments = Pass, item moves to TrainingApprovalQueue
[ ]	TrainingReviewQueue depth returns to 0
[ ]	TrainingApprovalQueue depth shows 1

STEP 6 — Approve the Document and Verify Workflow Completes (5 min)

27. In ACCE Process Configuration Console, expand TrainingApprovalQueue
28. Click on the work item
29. Set the following fields:

F_ApprovalStatus	`Approved`
F_ApprovedBy	`ceadmin`

30. Click Step / Dispatch to move the work item
31. The workflow should transition to the ApprovalNotification step, then MilestoneStep, then Terminate
32. In ACCE > Process Configuration Console, navigate to Completed Workflows
33. Verify DocumentApprovalWorkflow appears with status: Completed

[SCREENSHOT]
ACCE Process Configuration Console Completed Workflows list showing DocumentApprovalWorkflow instance with status Completed

	VERIFICATION CHECKLIST
[ ]	Work item stepped with F_ApprovalStatus = Approved
[ ]	TrainingApprovalQueue depth returns to 0
[ ]	Workflow instance appears in Completed Workflows with status: Completed
[ ]	Workflow instance shows milestone: ProcessComplete

Day 7 Lab Completion Summary

DocumentApprovalWorkflow is deployed and fully tested. Ready for event-driven automation.

Lab	Title	Skills	Status
LAB-07-01	Install & Connect Process Designer	IBM IM, PE connection, interface	[ ] Complete
LAB-07-02	Build Document Approval Workflow	Data fields, steps, routing, escalation	[ ] Complete
LAB-07-03	Deploy, Test & Verify Workflow	Deploy, queue stepping, completion	[ ] Complete